

K - Means Clustering

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K Means Clustering groups similar data points into clusters without needing labelled data.

It is used to uncover hidden patterns when the goal is to organise data based on similarity.

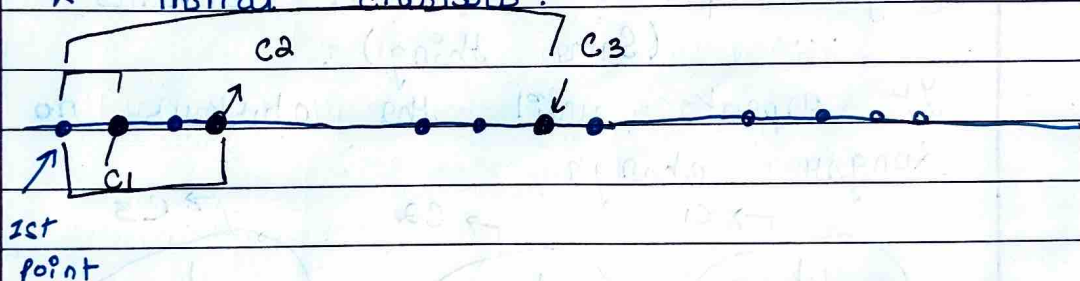
Steps

(Stat Quest)

Step 1: Select the no of clusters you want to identify in the data. This is the 'k' in k means clustering.

Step 2 Randomly select k distinct data points.

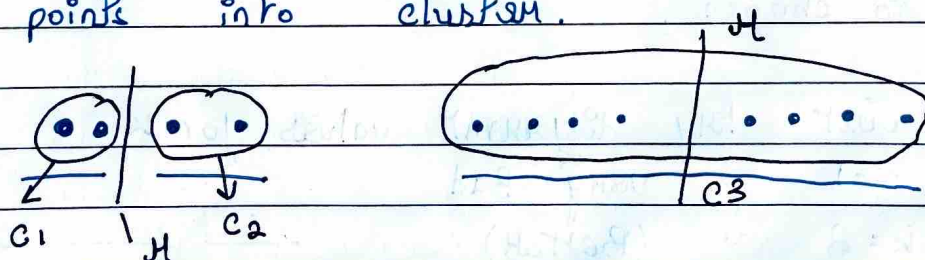
Step 3 Measure the distance b/w 1st point & the k initial clusters.



Step 4: Assign the first point to the nearest cluster.

Do the same thing for next point.

• Keep measuring the distance & assign points into cluster.



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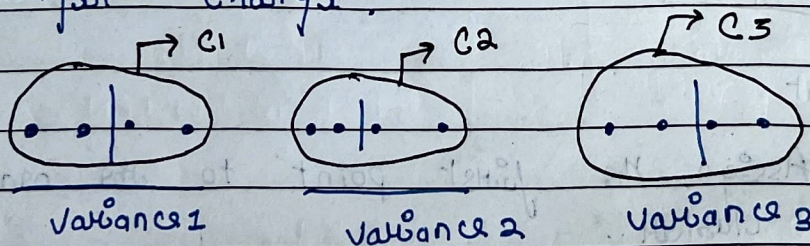
Steps: Calculate Mean of Each cluster & Repeat the same process.
• measure the distance & put the points into cluster.

Since k means clustering can't be the best clustering its only option is to keep track of these clusters & their total variance & do the whole thing over again with different starting points.

So, we start again, back at the Beginning

Again pick k initial clusters ...
(Same thing).

It repeats until the clusters no longer change.



Q) How to figure out what value of k to choose.

Sol Just try different values for k .

$k=1$ ~ very Bad

$k=2$ ~ (Better)

$k=3$ ~ (Even Better)

Quantify better by comparing the total variation within the 3 clusters to $k=3$.

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Elbow Method $\uparrow$ Variation $\downarrow$
(within each cluster)
Plot an Elbow plot. To plot k by finding Elbow in the plot.

Elbow Method : to find Best k .

Step 1: Apply k means clustering algo & form the clusters for different values of k .
Eg: $k = [1, 2, 3, 4, \dots]$

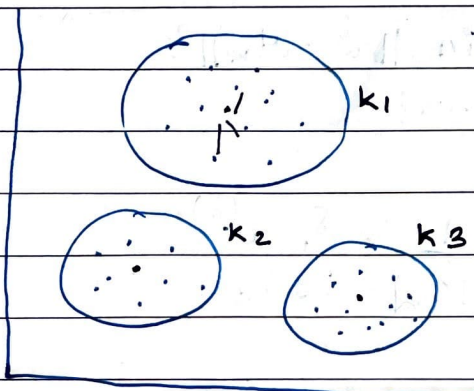
Step 2: For each k calculate the within cluster sum of squares (WCSS).

$$WCSS = \sum_{c_k} \left(\sum_{d_i \text{ in } c_k} \text{distance}(d_i, c_k)^2 \right)$$

where, $c \rightarrow$ cluster centroids &
 $d \rightarrow$ data point in each cluster.

OR

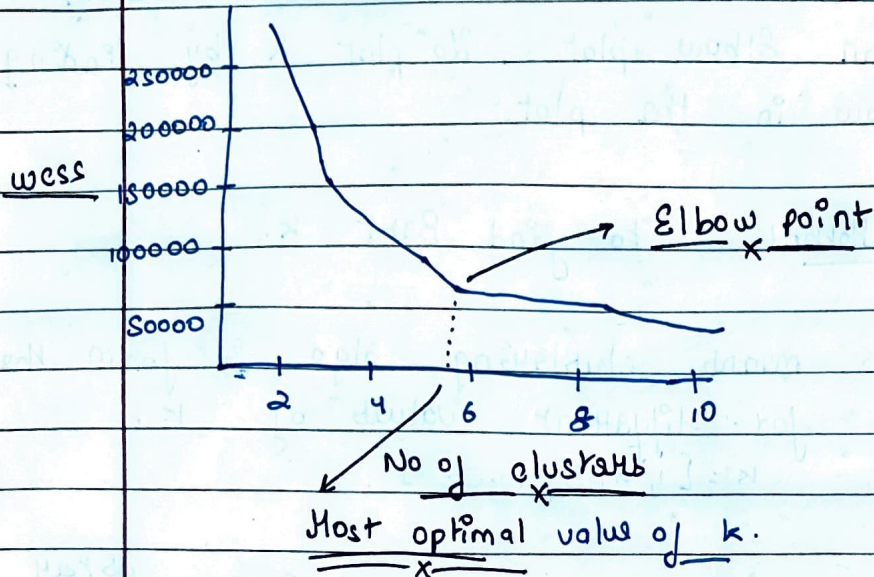
$$WCSS = \sum_{k=1}^K \left(\sum_{i=1}^n (x_i - \mu_k)^2 \right)$$



PRO $\rightarrow$

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Step 3 Plot curve of wcss according to the no of clusters.



#

KEY POINTS:

K Means clustering is highly sensitive to outliers. If data has outliers, then remove outliers or scale the data points.

K Means Mathematically

$$X = \{x^{(1)}, x^{(2)}, \dots, x^{(n)}\} \quad x^{(i)} \in \mathbb{R}^d$$

$$\mu_1, \mu_2, \mu_3, \dots, \mu_k \quad \xrightarrow{k \text{ Centroids}}$$

$$C_i = \arg \min_{k \in \{1, \dots, k\}} \|x^{(i)} - \mu_k\|^2$$

$$\mu_k^{(t+1)} = \frac{1}{|C_k|} \sum_{i \in C_k} x^{(i)}$$

$$J(\tilde{C}, \mu) = \sum_{j=1}^n \sum_{x^{(i)} \in C_j} \|x^{(i)} - \mu_j\|^2$$

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k-Means clustering - Numerical.

Initial Centroids:

$A_1 : (2, 10)$

$B_1 : (5, 8)$

$C_1 : (1, 2)$

Sol

	Data Points			Distances to.			cluster	
				2 10	5 8	1 2		
A_1	2	10						
A_2	2	5	0.0	3.61	8.00		3	New Cluster
A_3	8	4	5.0	4.24	3.16		2	
B_1	5	8	8.4	8.00	7.28		2	
B_2	7	5	3.6	0.00	7.21		2	
B_3	6	4	7.0	3.61	6.71		2	
C_1	1	2	7.2	4.12	5.39		3	
C_2	4	9	8.06	7.21	0.00		3	
			2.24	1.41	7.62		2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Now Centroids:

$A_1 : (2, 10)$

$B_1 : (6, 6)$

$C_1 : (1.5, 3.5)$

Same Process Repeat.

Now Centroids

$A_1 : (3, 9.5)$

$B_1 : (6.5, 5.25)$

$C_1 : (1.5, 3.5)$

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② Initial Centroids :

$M_1 : 4$

$M_2 : 11$

Data Points	Dist To:		cluster
	M_1	M_2	
2	2	9	$\rightarrow C_1$
4	0	7	$\rightarrow C_1$
10	6	1	$\rightarrow C_2$
12	8	1	$\rightarrow C_2$
3	1	8	$\rightarrow C_1$
20	16	9	$\rightarrow C_2$
30	26	19	$\rightarrow C_2$
11	7	0	$\rightarrow C_2$
25	21	14	$\rightarrow C_2$

$$d(x_2, x_1) = \sqrt{(x_2 - x_1)^2}$$

New Centroids

$$C_1 = \{2, 4, 3\}$$

$$M_1 = 3$$

$$C_2 = \{10, 12, 20, 30, 11, 25\}$$

$$M_2 = 18$$

			cluster	New cluster.
2	1	16	C_1	C_1
4	1	14	C_1	C_1
10	7	8	C_2	C_1
12	9	6	C_2	C_2
3	0	15	C_{21}	C_1
20	17	2	C_{22}	C_2
30	27	12	C_2	C_2
11	8	7	C_2	C_2
25	22	7	C_2	C_2

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New centroid :

$$M_1 = 4.75$$

$$M_2 = 19.6$$

Data Points	Dist to		cluster	New cluster.
	M_1	M_2		
2	2.7	17.6	C_1	C_1
4	0.7	18.6	C_1	C_1
10	5.25	9.6	C_1	C_1
12	7.25	7.6	C_2	C_1
3	1.75	16.6	C_1	C_2
20	18.25	0.4	C_2	C_2
30	28.25	10.4	C_2	C_1
11	6.25	8.6	C_2	C_2
25	20.25	5.4	C_2	

Now CMM

$$M_1 : 7$$

$$M_2 : 25$$

Same

Final cluster are:

$$C_1 = \{2, 4, 10, 11, 12, 3\}$$

$$C_2 = \{20, 30, 25\}$$